

MECH 335 Project Automation



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1.0 Abstract

The following report outlines the design of an automaton that illustrates a moment in a sports event. The purpose of this course project was to further understand and apply the theoretical principles regarding gears, cam gears, and four-bar mechanisms learned in class. Our group consisted of four people and we each developed a concept to fulfill the project design requirements. Concept 1 (World Cup Goal) tells the story of a free kick at a world cup. Concept 2 (Sport Fisherman) illustrates a fisherman catching a fish while there are waves, birds, and fishes moving. Concept 3 (Baseball Hit) shows a batter hitting a ball when pitched and a runner going to first base. Concept 4 (An Olympic swimmer with a flying fish) tells the story of a swimmer that raced a flying fish in the last two laps of a ten kilometer race in the Tokyo 2020 olympic games.

By constructing a weighted objectives table with a ranking matrix and appropriate score criterion decided by the design objectives, we concluded that the most feasible design was the World Cup Goal. It was not complicated and was inexpensive when compared to other designs. The automaton is composed of two four bar mechanisms, a cam gear, a belt, and a few spur gears. A crank is rotated which drives the main gear which is connected to the four bar linkage that drives the motion of the soccer ball. This same gear drives an intermediate gear shaft system which was implemented to avoid interference between the ball and the main shaft. This intermediate shaft has a gear at the other end which drives yet another gear which is the main driver for the cam mechanism. The cam mechanism mimics the crowd cheering as the ball goes into the net. The main shaft also has a belt which connects to another shaft that is the main driver for the leg kicking four bar mechanism. We chose a belt as it seemed to be the easiest to implement and maintained the same speed of the shafts.

By picking the precision points required for the soccer player's legs as well as the ball, we were able to synthesize the four bar mechanisms through both graphical and analytical methods. We were able to recreate the four bar mechanism in SolidWorks using the lengths obtained through the analysis and verify the design using Working Model. Reflecting on the design, we could have obtained smoother motions for the four bar mechanisms through multiple tests and iterations. A more in depth analysis of gear ratio and speeds could have led to an improvement in synchronization and could reduce the size of the automaton. One of the challenges faced during the design process was that the spatial cylinder cam initially chosen was too complex for the desired motion. The Scotch Yoke mechanisms initially designed for the crowd movement to mimic the cheering was difficult to implement as it did not provide adequate control and synchronization with the other mechanical elements of the automaton. As a solution, we transitioned to a gear-driven system. Through a set of gears we were able to implement more precise timing and coordination between elements.

2.0 Introduction

Our course project challenges us to design an automation that simulates a moment from a sports event. An automation is a machine designed to follow a predetermined set of operations automatically. In this context, we're tasked with creating a mechanical representation of a sports scene, which should be both functional and visually representative of the event.

The core of our project involves the design and analysis of two main mechanisms. These mechanisms will be the building blocks of our automaton, enabling it to mimic the dynamic actions of a sports event through mechanical movement. We will be working with four-bar linkages, which must be designed to meet specific motion requirements. These linkages will create the primary actions of our automaton, with additional mechanical elements such as cams and gears integrated to enhance the complexity and realism of the motion.

Each team member will contribute a design concept, which will be evaluated and selected based on a set methodology to ensure the most effective design is chosen for development. Our final design will be developed through CAD models, and the functionality will be demonstrated through a simulation in the Working Model software. This report will document our design process, from initial concepts to the synthesis of the final mechanisms, reflecting the analytical and practical skills applied throughout the project.

3.0 Design Concepts

3.1 World Cup Goal, Luca Carinelli

The "World Cup Goal" concept captures the climactic moment of a soccer match during the World Cup. This automaton will recreate the excitement of a player scoring a goal, with the crowd's reaction. As the crank is turned, the observer will see a soccer ball being 'kicked' into a net and the spectators in the stands rising in excitement.

Four-bar Linkage for the Kicking/Ball Motion: Two four-bar linkages are used to simulate the kicking action of a player and the ball flight path. The linkages ensure that the motion is realistic, with the proper trajectory as the player's leg strikes the ball. The linkage's movement is timed to coincide with the ball's apparent movement towards the goal. The dyad of both linkages are moved using a Scotch Yoke system (blue).

Scotch Yoke/Cam for Spectator Movement: The crowd's reaction is animated using a Scotch Yoke and a Cam, which converts rotational motion into linear motion. This system causes the spectators to rise from their seats in unison, reflecting the sudden burst of excitement as the goal is scored.

Integrated Crank: The entire system is driven by a single crank that, when turned, moves a Scotch Yoke system (orange) that coordinates all movements to simulate the scoring of a goal. This crank mechanism is the heart of the automaton, ensuring all movements are synchronized and fluid.

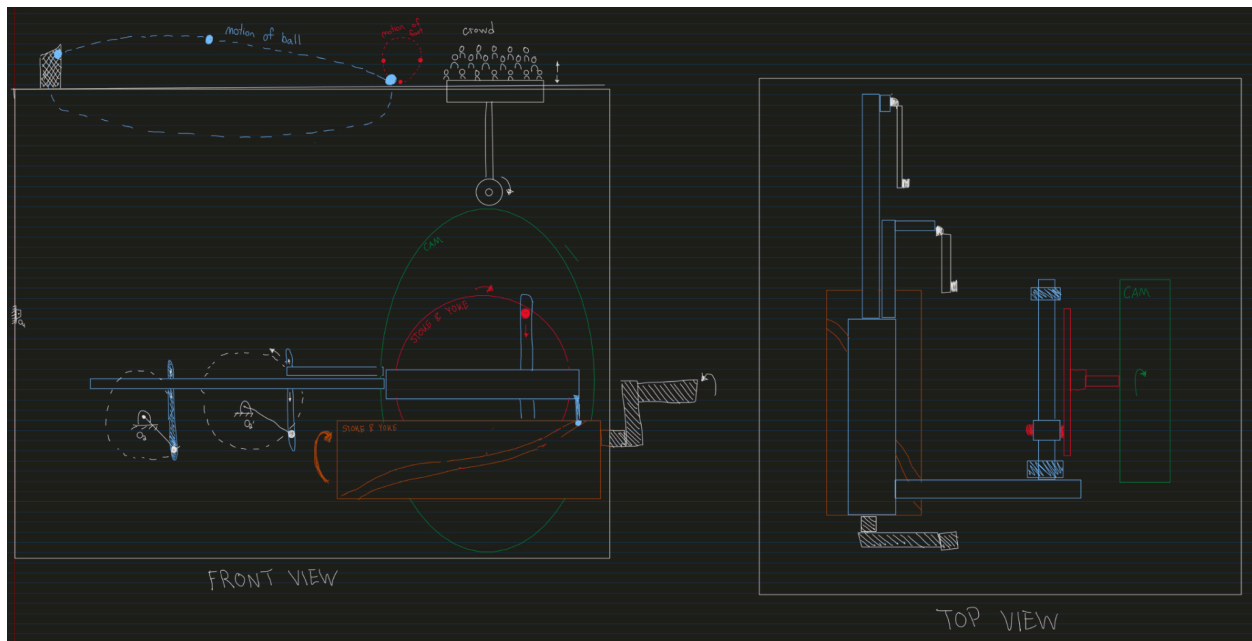


Figure 1: World Cup Goal Concept Design Sketch

3.2 Sport Fisherman, by Noah Magee

The figure to the right illustrates the general concept for the “Sport Fisherman” automaton. This concept features a fisherman who is trying to reel in a fierce catch, while a seabird flies in the background. The design also attempts to simulate the movement of the ocean, through the oscillation of a “wave plate” rocker. This design captures the excitement of catching a fish, while at the same time, incorporates the natural movement of our oceanic environment.

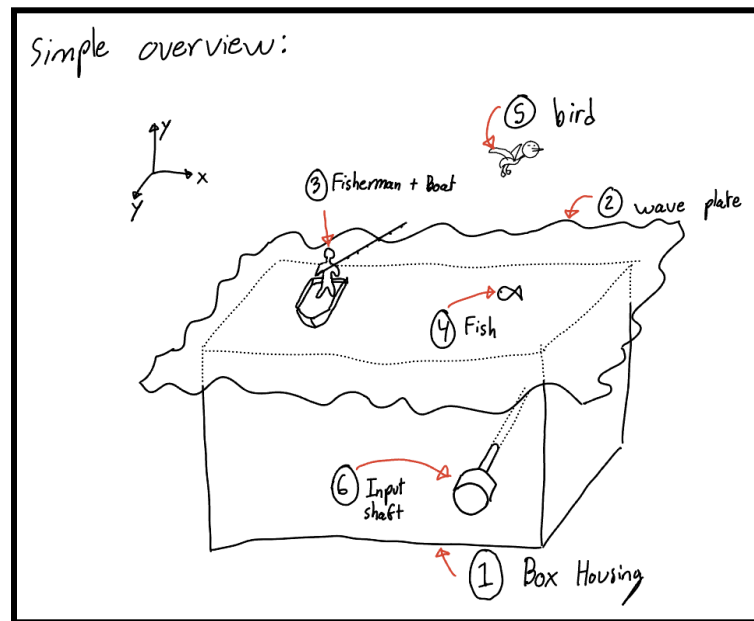


Figure 2: Sport Fisherman Concept Design Sketch

Four Bar linkage for the fish’s motion: The fish’s motion is defined by 3 positions (Motion Generation). This mechanism will be synthesized through the graphical method, as well as the Dyad Method (both with specified ground pivots). One of the ground pivots will be connected directly to the ‘input shaft’, as seen in the figures on the following page.

Four bar linkage for the Bird’s motion: The bird will be positioned at the top-most point of the triangle link, seen in the figures on the following page. The bird’s motion will be synthesized both graphically, and analytically, with the bird passing through 3 precision points (Path Generation). This four bar linkage will be powered by the ‘input shaft’.

Rod Motion: The Fisherman’s rod will move up and down, to imitate how a fisherman would reel in a fish. In order to make the motion a little more realistic, the rod will be driven by a ‘fast return mechanism’ which will make the rod tip raise slower than it is lowered. The fast return mechanism will be powered by a belt, which will be attached directly to the ‘input shaft’.

Wave Plate Motion (rocker): The ‘wave plate’ is a solid plate made to look like the surface of the ocean. As I only want this plate to move slightly, I will treat the plate as a rocker, and use an

additional dyad to 'rock' the plate back and forth. The Dyad will be connected through a series of gears to the 'input shaft'. As seen in Figure ____, the 'Wave Plate' has 2 'Link 2's, as well as 2 'link 4's. This is simply done for stability of the plate, and does not affect the motion. This does not affect the motion, as the respective links coincide when looking from the 'XY-plane' view (view figure __).

Movement Frequency: The 'rod' and the fish's movement will be synced up, giving the impression that the fish is being 'tugged' by the rod. To accomplish this, the fish and the rod will use the same input shaft to power their movement, allowing them to have the same movement frequency. The 'Wave Plate' motion on the other hand, will have a slower movement frequency, as waves typically move pretty slow. To accomplish this, a series of gears will be used to reduce the angular velocity of the 'input shaft' to allow for a slow rotating dyad, and therefore the plate will slowly rock back and forth.

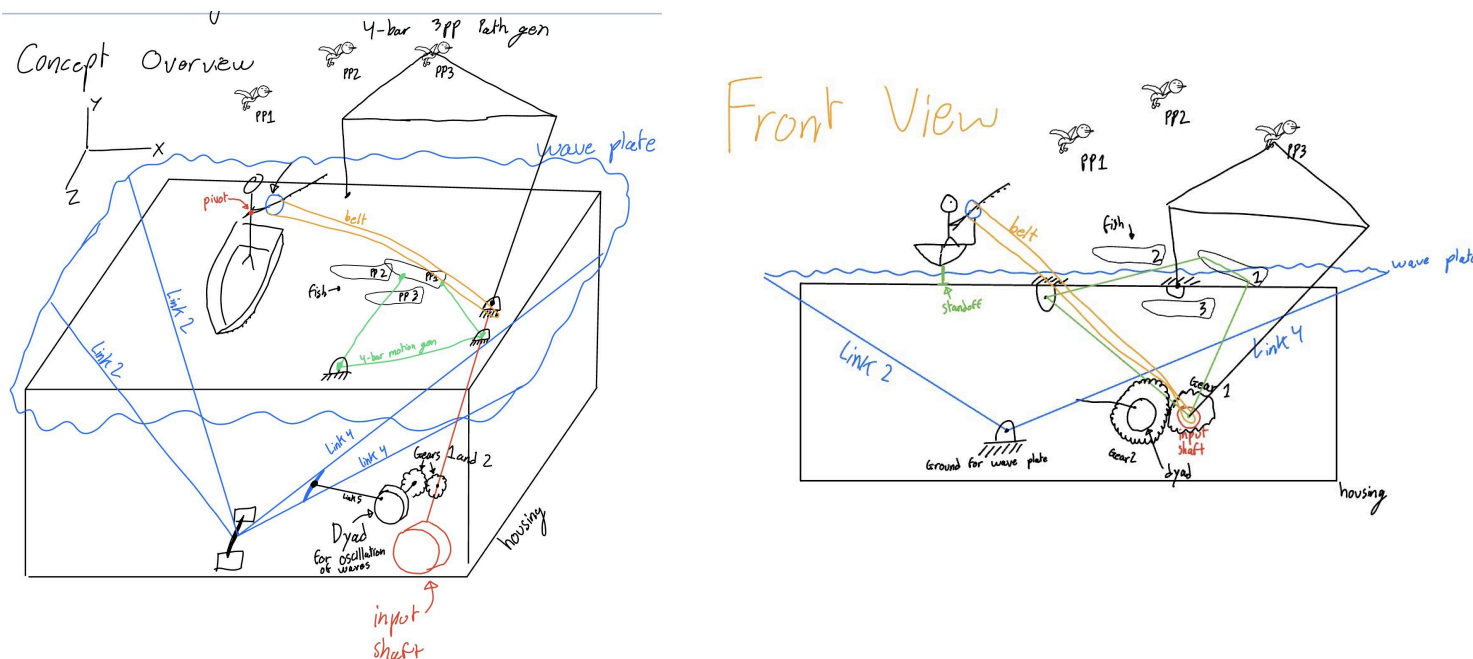


Figure 3: Sport Fisherman Concept Design Sketch

3.3 Baseball Hit, Rodrigo Padres

The figure below illustrates the concept of a baseball home run. The automaton is composed of an input shaft, a worm gear, some more spur gears, a cam and two four bar linkages. There is a batter, the crowd, the pitcher, basemen, and a runner. The design illustrates a baseball hit with one runner going into second base.

Four-bar Linkage for the baseball and runner: One of the four-bar mechanisms will be attached to the baseball and three precision points will be used to mimic a hit into one of the outfield players gloves. The mechanism will be driven by a combination of spur gears and the input shaft. The timing and ratio needed will be determined. The second four-bar mechanism drives the runner from first to second base. It will be driven in a similar way to the other four-bar mechanism.

Cams/Gears used for Crowd Cheering and Hitter: The design will also use a cam and spur gears to mimic a crowd cheering in the stands. The hitter will also be swinging at the baseball through the use of a worm gear which is essentially driven by a spur gear that will be attached to the input shaft.

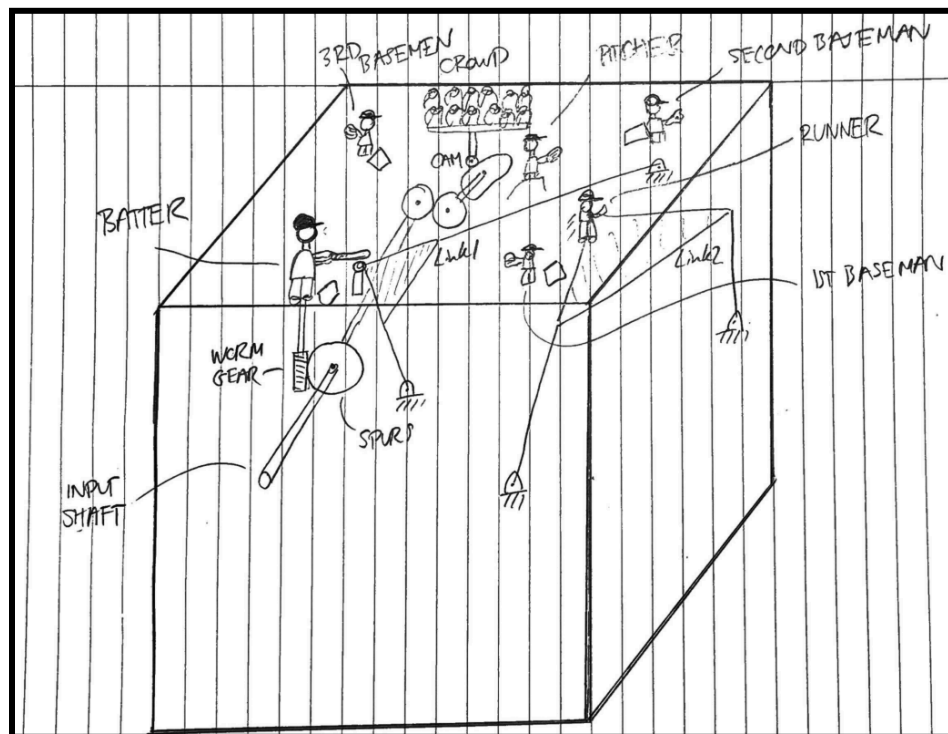


Figure 4: Baseball Hit Concept Design Sketch

3.4 An Olympic swimmer with a flying fish , Devin Woudenberg

The figure below tells a story of a swimmer in the 2020 Tokyo Olympics that was racing a flying fish in the last two laps of a ten kilometer race. The fish did not end up winning but nonetheless it was a great story from those olympic games.

Four-bar Linkage for the swimmers arms, and the piranha: The two four-bar linkages will be used for the swimmer's arms to help him propel himself forward away from the piranha. The second four-bar mechanism will be used to show the motion of the piranha moving towards the swimmer. The two four bar linkages will be driven by an assembly of gear connected to the input shaft.

Cams/Gears used for leg movement of the swimmer: The two other families of mechanisms will be using cams and gears to reproduce the movement of legs kicking in water. Their will be two cams offset by a specified angle so that they move oppositely of each other (ie. when one leg moves up the other moves down). The gears will be connected by a drive shaft that is turned by the input shaft of the automaton.

Conclusion: This automaton will use material learned in MECH 335. It will put the theory learned into a practical use exhibiting the use of four-bar linkages, cams, and gears. It will be run using a singular input crank that will allow for all the mechanisms to move simultaneously.

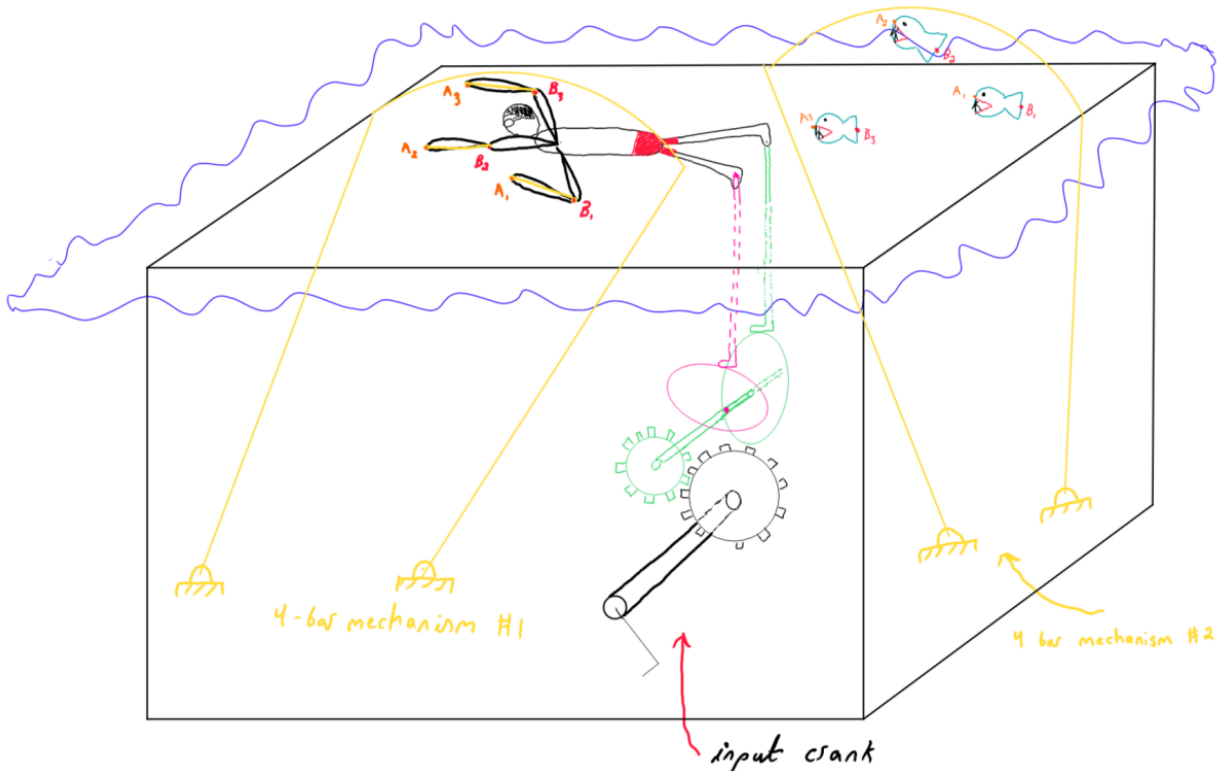


Figure 5: Olympic swimmer versus a flying fish

4.0 Design Selection

The design was chosen by evaluating each concept using a weighted selection table. The design objectives for the criteria were chosen to be manufacturability, cost estimate, size, number of mechanism families, and the estimated design hours. In the ranking matrix, we specify which criteria is more important. For example, The Cost Estimate criteria was considered to be more important than the manufacturability and thus on the left side of the matrix a value of 1 was assigned to the horizontal Cost Estimate row.

The design chosen as a result of the chart is the World Cup Goal concept. It earned the highest numeric score at 8.2 followed by the Swimming Piranha at 7.9. The Baseball Hit concept had a score of 7.2 and the Sport Fisherman had a score of 6. The criteria used to evaluate each concept is shown by the tables below.

Table 1.1 Manufacturability Criterion

Manufacturability	Num. Score
Simple/Easy	10
Some Complex	6
Mostly Complex Parts	4
All Complex	0

Table 1.2 Cost Estimate Criterion

Cost Estimate	Num. Score
< \$ 100	10
\$100-200	8
\$200-300	6
\$300-400	4
>\$400	0

Table 1.3 Size Estimate Criterion

Size	Num. Score
0-700 cm ³	10
700-900 cm ³	8
900-1100 cm ³	5
> 1100 cm ³	2

Table 1.4 Families of Mechanisms Criterion

Families of Mechanisms	Num. Score
-------------------------------	-------------------

> 4	4
4	3
3	2
2	1

Table 1.5 Estimated Design Hours Criterion

Design Hours	Num. Score
0-2	10
2-4	9
4-6	8
6-8	4
> 8	2

Table 1.6 Ranking Matrix

Criteria	Manufacturability	Cost Estimate	Size	Families of Mechanisms	Design Hours	Row Total	Normalized Weights
Manufacturability	n.a.	0	1	1	1	3	0.3
Cost Estimate	1	n.a.	0	1	0	2	0.2
Size	0	1	n.a.	0	0	1	0.1
Families of Mechanisms	0	0	1	n.a.	0	1	0.1
Design Hours	0	1	1	1	n.a.	3	0.3

Table 1.7 Weighted Selection Table

			Design Options							
			Concept 1: World Cup Goal		Concept 2: Sport Fisherman		Concept 3: Baseball Hit		Concept 4: Swimmer vs. Flying fish	
Design Objective	Rel. Weight	Evaluation Scale	Numeric Score	Weighted Value	Numeric Score	Weighted Value	Numeric Score	Weighted Value	Numeric Score	Weighted Value
Manufacturability	0.3	Table 1.1	10	3	6	1.8	10	3	10	3

Cost Estimate	0.2	Table 1.2	8	1.6	6	1.2	6	1.2	8	1.6
Size	0.1	Table 1.3	8	0.8	5	0.5	5	0.5	8	0.8
Families of Mechanisms	0.1	Table 1.4	1	0.1	1	0.1	1	0.1	1	0.1
Design Hours	0.3	Table 1.5	9	2.7	8	2.4	8	2.4	8	2.4
			Total:	8.2	Total:	6	Total	7.2	Total	7.9

Concepts 1, 3 and 4 were given a manufacturability score of 10 because we do not foresee the concepts having any complicated manufactured parts. The sports fisherman, however, might have complications with the wave plate as it might interfere with other moving parts. Therefore, it was assigned a score of 6.

Concepts 1 and 4 had the simplest designs and least amount of moving components which would result in a low cost. They were assigned a cost of 8. Concepts 2 and 3 had more components which would result in higher costs. They were assigned a score of 6. The size scores were also estimated by the amount of components. Therefore Concepts 1 and 4 had a score of 8 while the others a score of 5.

The score for families of mechanisms was found to be the same across all concepts. This was because each concept only accounted for the minimum design requirements as illustrated in the project outline. The estimated design hours were estimated by considering how complicated the timing of the “story” would be, as well as the amount of components and their complexity. Because all of the designs meet the basic design requirement they are all not very complicated. The World Cup Goal concept appeared to be the least expensive alternative as there was already a clear idea of how the linkages and gears would interact. Therefore, it was assigned a score of 9 while the rest a score of 8.

5.0 Detailed Description

This display is set into motion by a central crank mechanism that serves as the heart of the automaton, initiating movement through a sequence of meticulously arranged mechanical components.

The crank is directly coupled to a 'main gear' which is the primary distributor of motion to the rest of the automaton. This gear's strategic position allows it to transfer motion to two other gears and a belt, each integral to the operation of three separate yet harmoniously functioning mechanisms.

On the left side of the main gear, a larger gear with an increased diameter provides a deliberate deceleration of motion, critical for the realistic portrayal of the soccer ball's trajectory. This gear is the driver for the first four-bar linkage mechanism, which dictates the path of the ball. Through this arrangement, the ball is propelled in an arcing motion that mimics the parabolic flight typical of a well-kicked soccer ball, terminating in its eventual contact with the net.

Concurrently, the main gear also imparts motion to an identically sized gear on its right. This gear translates rotational motion to a cam mechanism that is responsible for animating the crowd of fans. The cam is created to synchronize the spectators' movement with the ball's arrival in the net, creating an instant of collective celebration.

Completing the trio of animated components is the four-bar linkage mechanism that constitutes the soccer player's leg. The motion is transferred from the crank to this mechanism via a vertical belt or chain drive, which effectively converts the rotational motion vertically upwards. This linkage is engineered to simulate the leg drawing back before striking the ball, delivering a powerful and accurate representation of a soccer kick.

6.0 Precision Points

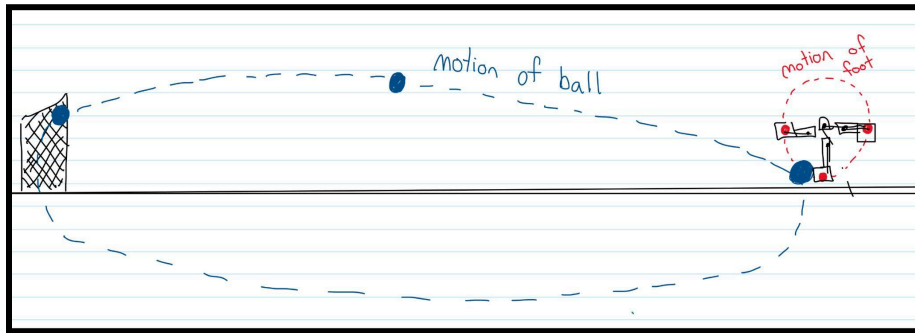


Figure 6: General Idea of Precision Points for Both Four Bar Mechanisms

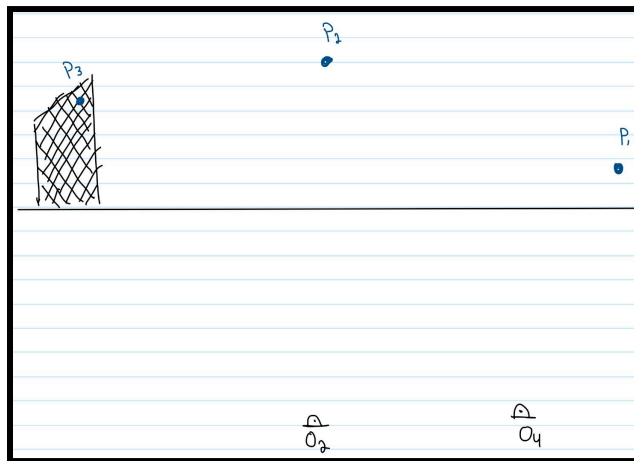


Figure 7: Precision Points and Ground Pivots for Ball Mechanism

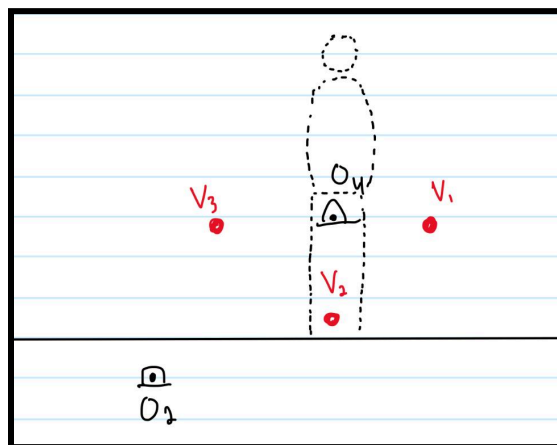


Figure 8: Precision Points and Ground Pivots for Foot Mechanism

5.0 Synthesis of Mechanisms

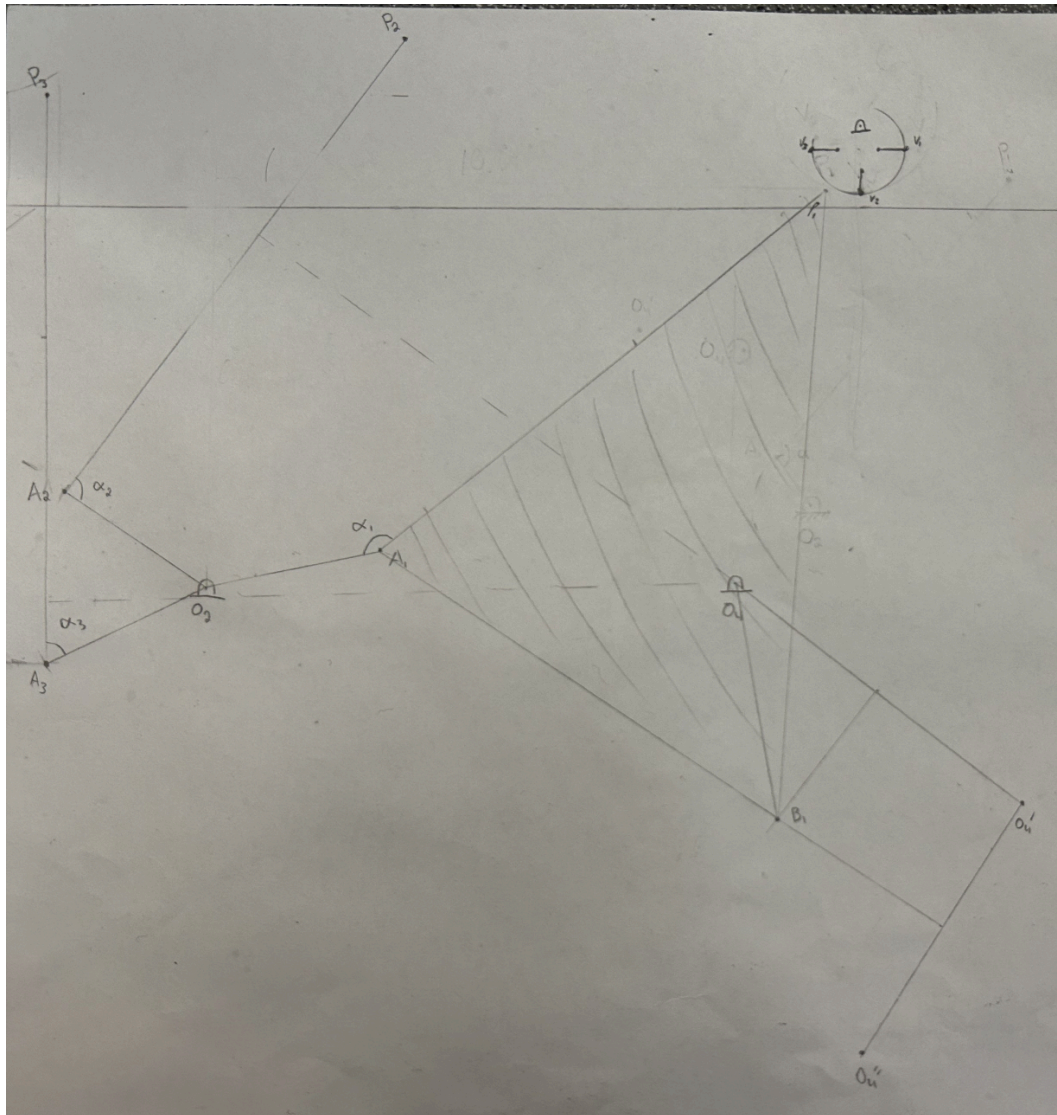


Figure 9: Graphical Synthesis of Four Bar Ball Mechanism - Path Generation, Three Precision Points (Coordinate Frame Approach)

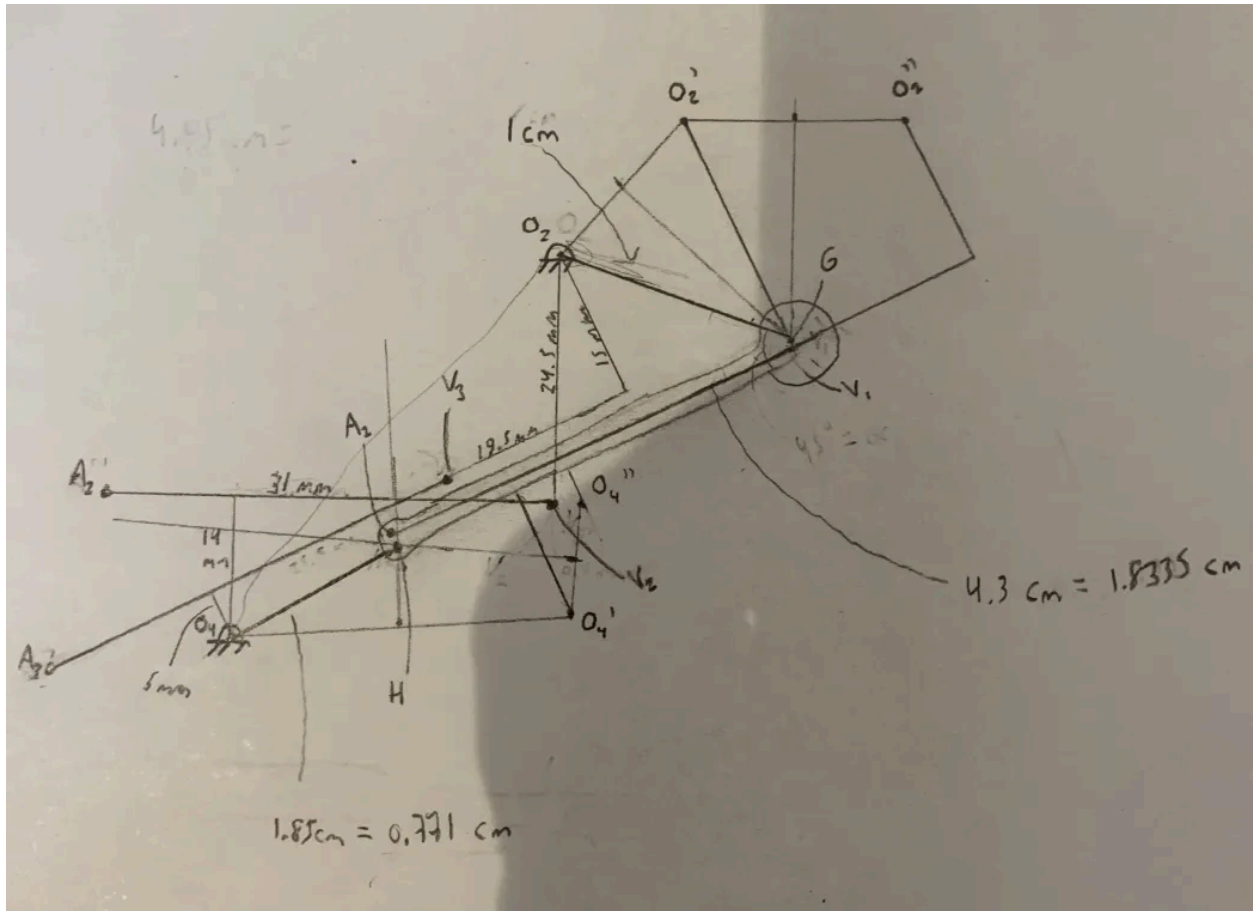


Figure 10: Graphical Synthesis of Four Bar Foot Mechanism - Motion Generation, Three Precision Points (Coordinate Frame Approach)

$$R_1^{(1)} = 10.5 + 6.6i$$

Prescribed timing
(From graphical method) $\rightarrow \beta_2 = 133^\circ \quad \beta_3 = 193^\circ$

LHS

$$\delta_2 = -7.3 + 2.6i$$

$$\delta_3 = -13.4 + 1.5i$$

$$D_1 = (R_1^{(1)} + \delta_3)e^{i\beta_2} - (R_1^{(1)} + \delta_2)e^{i\beta_3}$$

$$D_2 = -(R_1^{(1)} + \delta_3) + R_1^{(1)}e^{i\beta_3}$$

$$D_3 = R_1^{(1)} + \delta_2 - R_1^{(1)}e^{i\beta_2}$$

subbing in values:

$$D_1 = (10.5 + 6.6i - 13.4 + 1.5i) [\cos(133) + \sin(133)i] - (10.5 + 6.6i - 7.3 + 2.6i) [\cos(193) + \sin(193)i]$$

$$D_1 = -2.89773 + 2.03893i$$

$$D_2 = -(10.5 + 6.6i - 13.4 + 1.5i) + (10.5 + 6.6i) [\cos(193) + \sin(193)i]$$

$$D_2 = -5.84620 - 16.89282i$$

$$D_3 = 10.5 + 6.6i - 7.3 + 2.6i - (10.5 + 6.6i) [\cos(133) + \sin(133)i]$$

$$D_3 = 15.18791 + 6.02197i$$

$$\alpha_2 = 2\arg(-D_1) - \arg(D_2) - \arg(D_2 e^{i\beta_2})$$

$$\alpha_2 = 2 \tan^{-1} \left[\frac{-2.03893}{+2.87773} \right] - \tan^{-1} \left(\frac{-16.89282}{-5.84620} \right) - \tan^{-1} \left[\frac{16.34172 + 7.24523i}{-5.84620 - 16.89282i} \right] (\cos(93) + \sin(93)i)$$

324.87° 250.91° 23.91°

$$\alpha_2 = 2(324.87) - 250.91 - 23.91$$

$$= 374.92 \Rightarrow 374.92 - 360 = 14.92$$

$$\alpha_3 = 2\arg(-D_1) - \arg(D_3) - \arg(D_3 e^{i\beta_3})$$

$$= 2 \tan^{-1} \left(\frac{-2.03893}{+2.87773} \right) - \tan^{-1} \left(\frac{6.02197}{15.18791} \right) - \tan^{-1} \left[\frac{5.18791 + 6.02197i}{(5.18791 + 6.02197i)(\cos(193) + \sin(193)i)} \right]$$

324.87° 21.63° 214.62°

$$\alpha_3 = 2(324.87) - 21.63 - 214.62$$

$$= 413.49 \Rightarrow 413.49 - 360 = 53.49$$

repeating values $\Rightarrow \beta_2 = 133^\circ \quad \beta_3 = 193^\circ$

$$\delta_2 = -7.3 + 2.6i$$

$$\delta_3 = -13.4 + 1.5i$$

$$h_1^{(1)} = 10.5 + 6.6i$$

$$\alpha_2 = 14.92^\circ$$

$$\alpha_3 = 53.49^\circ$$

$$R_2 = \frac{\begin{vmatrix} \delta_2 & e^{i\alpha_2} - 1 \\ \delta_3 & e^{i\alpha_3} - 1 \end{vmatrix}}{\begin{vmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 \end{vmatrix}}} = \frac{\delta_2(e^{i\alpha_3} - 1) - \delta_3(e^{i\alpha_2} - 1)}{(e^{i\beta_2} - 1)(e^{i\alpha_3} - 1) - (e^{i\beta_3} - 1)(e^{i\alpha_2} - 1)}$$

$$= \frac{\delta_2(\cos\alpha_3 + \sin\alpha_3 i - 1) - \delta_3(\cos\alpha_2 + \sin\alpha_2 i - 1)}{(\cos\beta_2 + \sin\beta_2 i - 1)(\cos\alpha_3 + \sin\alpha_3 i - 1) - (\cos\beta_3 + \sin\beta_3 i - 1)(\cos\alpha_2 + \sin\alpha_2 i - 1)}$$

$$= \frac{0.80145 - 3.41982i}{(0.09344 - 1.64813i) - (0.12448 - 0.50075i)} = 2.95948 + 0.77856i$$

$$R_3' = \frac{\begin{vmatrix} e^{i\beta_2} - 1 & \delta_2 \\ e^{i\beta_3} - 1 & \delta_3 \end{vmatrix}}{\begin{vmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 \end{vmatrix}}} = \frac{(\cos\beta_2 + \sin\beta_2 i - 1)(\delta_3) - (\cos\beta_3 + \sin\beta_3 i - 1)(\delta_2)}{(\cos\beta_2 + \sin\beta_2 i - 1)(\cos\alpha_3 + \sin\alpha_3 i - 1) - (\cos\beta_3 + \sin\beta_3 i - 1)(\cos\alpha_2 + \sin\alpha_2 i - 1)}$$

$$= \frac{(\cos\beta_2 + \sin\beta_2 i - 1)(\delta_3) - (\cos\beta_3 + \sin\beta_3 i - 1)(\delta_2)}{(0.09344 - 1.64813i) - (0.12448 - 0.50075i)} = 7.54 + 5.82i$$

RHS

$$B_1^{(2)} = 1.3 + 6.7i$$

$$D_1 = (B_1^{(2)} + \delta_3)(\cos \alpha_2 + \sin \alpha_2 i) - (B_1^{(2)} + \delta_2)(\cos \alpha_3 + \sin \alpha_3 i)$$

$$= -2.75863 + 4.09752i$$

$$D_2 = -(B_1^{(2)} + \delta_3) + (B_1^{(2)})(\cos \alpha_3 + \sin \alpha_3 i)$$

$$= 7.48830 - 3.16886i$$

$$D_3 = (B_1^{(2)}) + \delta_2 - (B_1^{(2)})(\cos \alpha_2 + \sin \alpha_2 i)$$

$$= -5.53112 + 2.49117i$$

$$\delta_2 = -7.3 + 2.6i$$

$$\delta_3 = -13.4 + 1.5i$$

$$\alpha_2 = 14.92^\circ$$

$$\alpha_3 = 93.49^\circ$$

$$\gamma_2 = 2 \tan^{-1} \left(\frac{-4.09752}{2.75863} \right) - \tan^{-1} \left(\frac{-3.16886}{7.48830} \right) - \tan^{-1} \left[\underbrace{(7.48830 - 3.16886i)(\cos \alpha_2 + \sin \alpha_2 i)}_{8.05172 - 1.13401i} \right]$$

$$\gamma_2 = 2(303.95) - 337.06 - 351.98$$

$$= -81.14 \Rightarrow \boxed{278.86^\circ}$$

$$8.05172 - 1.13401i$$

$$\Downarrow$$

$$\frac{-1.13401}{8.05172}$$

$$\gamma_3 = 2 \tan^{-1} \left(\frac{-4.09752}{2.75863} \right) - \tan^{-1} \left(\frac{+2.49117}{-5.53112} \right) - \tan^{-1} \left[\frac{(-5.53112 + 2.49117i)(\cos \alpha_3 + \sin \alpha_3 i)}{\dots} \right]$$

$\angle -56.05^\circ$ $180 + (-24.25) = 155.75^\circ$

$$\gamma_3 = 2(303.95) - 155.75 - 209.24$$

$$\gamma_3 = \boxed{242.91}$$

$180 + 27.24 = 207.24^\circ$ $-5.29309 - 2.96350i$
 $\Downarrow$
 $\frac{-2.96350}{-5.29309}$

$$h_4 = \frac{\begin{vmatrix} \delta_2 & e^{i\alpha_2-1} \\ \delta_3 & e^{i\alpha_3-1} \end{vmatrix}}{\begin{vmatrix} e^{i\alpha_2-1} & e^{i\alpha_3-1} \\ e^{i\alpha_2-1} & e^{i\alpha_3-1} \end{vmatrix}} = \frac{(\delta_2)(\cos \alpha_3 + \sin \alpha_3 i - 1) - (\delta_3)(\cos \alpha_2 + \sin \alpha_2 i - 1)}{(\cos \alpha_2 + \sin \alpha_2 i - 1)(\cos \alpha_3 + \sin \alpha_3 i - 1) - (\cos \alpha_3 + \sin \alpha_3 i - 1)(\cos \alpha_2 + \sin \alpha_2 i - 1)}$$

$$= \frac{0.80145 - 3.41982i}{(1.13681 - 0.27975i) - (0.27829 - 0.34470i)} = \boxed{0.62857 - 4.03094i}$$

$$h_3 = \frac{\begin{vmatrix} e^{i\alpha_2} & \delta_2 \\ e^{i\alpha_3} & \delta_3 \end{vmatrix}}{\begin{vmatrix} e^{i\alpha_2} & e^{i\alpha_3} \\ e^{i\alpha_2} & e^{i\alpha_3} \end{vmatrix}} = \frac{(\delta_3)(\cos \alpha_2 + \sin \alpha_2 i - 1) - (\delta_2)(\cos \alpha_3 + \sin \alpha_3 i - 1)}{(\cos \alpha_2 + \sin \alpha_2 i - 1)(\cos \alpha_3 + \sin \alpha_3 i - 1) - (\cos \alpha_3 + \sin \alpha_3 i - 1)(\cos \alpha_2 + \sin \alpha_2 i - 1)}$$

$$= \boxed{0.67101 + 10.73059i}$$

$$\begin{aligned} \gamma_2 &= 278.86^\circ \\ \gamma_3 &= 242.91^\circ \\ \delta_2 &= -7.3 + 2.6i \\ \delta_3 &= -13.4 + 1.5i \\ \alpha_2 &= 14.92^\circ \\ \alpha_3 &= 93.49^\circ \end{aligned}$$

Figure 11: Analytical Synthesis of Four Bar Ball Mechanism - Standard Dyad Method

LS:

$$\delta_2 = -1.0734 - 0.2614i$$

$$\delta_3 = -1.4089 - 0.5999i$$

$$R_1^{(1)} = 1.4796 + 0.8326i$$

$$R_2^{(1)} = 0.1458 - 0.7412i$$

$$\alpha_2 = -28.84$$

$$\alpha_1 = 0$$

$$D_1 = (R_1^{(1)} + \delta_3)e^{i\alpha_2} - (R_2^{(1)} + \delta_2)e^{i\alpha_1}$$

$$= (0.0707 + 0.2327i)(\cos(-28.84) + i\sin(-28.84)) - (0.9062 + 0.5712i)(1)$$

$$= (0.1742 + 0.1697i) - (0.9062 + 0.5712i)$$

$$D_1 = -0.732 - 0.4015i$$

$$D_2 = -(R_1^{(1)} + \delta_3) + R_2^{(1)}e^{i\alpha_3}$$

$$= -0.0707 - 0.2327i + (1.4796 + 0.8326i)$$

$$D_2 = 1.4089 + 0.5999i$$

$$D_2 = R_1^{(1)} + J_2 - R_1^{(1)} e^{i\alpha_2}$$

$$= 0.4062 + 0.5712i - 1.6977 + 0.0156i$$

$$D_2 = -1.2915 + 0.5556i$$

$$\beta_2 = 2 \tan^{-1}(-D_1) - \tan^{-1}(D_2) - \tan^{-1}(D_2 e^{i\alpha_1})$$

$$\beta_2 = 102.68^\circ$$

$$\beta_3 = 2 \tan^{-1}(-D_1) - \tan^{-1}(D_1) - \tan^{-1}(D_3 e^{i\alpha_3})$$

$$\beta_3 = 166.52^\circ$$

Cramer's Rule:

$$R_2 = \frac{\begin{vmatrix} \delta_2 & e^{i\alpha_2} - 1 \\ \delta_3 & e^{i\alpha_3} - 1 \end{vmatrix}}{\begin{vmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 \end{vmatrix}} = \frac{-2.91987 - 0.35121i}{-3.6966 + 1.04314i} = 0.70678 + 0.29445i$$

$$R_3 = \frac{\begin{vmatrix} e^{i\beta_2} - 1 & J_2 \\ e^{i\beta_3} - 1 & J_3 \end{vmatrix}}{\begin{vmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 \end{vmatrix}} = \frac{0.53671 - 0.7916i}{-3.6966 + 1.04314i} = -0.19095 + 0.16025i$$

R_5 :

$$D_1 = (R_2' + S_3)e^{i\omega_2} - (R_2^{(1)} + S_2)e^{i\omega_3}$$

$$D_1 = -0.8257 + 0.4371i$$

$$D_2 = -(R_2^{(1)} + S_2) + R_2^{(1)}e^{i\omega_3}$$

$$D_2 = 1.4089 + 0.5799i$$

$$D_3 = R_1 + S_2 - R_1e^{i\omega_2}$$

$$D_3 = -0.6978 - 0.2870i$$

$$\gamma_2 = 2\arg(-D_1) - \arg(D_2) - \arg(D_2e^{i\omega_2})$$

$$\gamma_2 = 73.08^\circ$$

$$\gamma_3 = 2\arg(-D_1) - \arg(D_3) - \arg(D_3e^{i\omega_3})$$

$$\gamma_3 = 99.96^\circ$$

CRAMER'S RULE:

$$R_4 = \frac{\begin{vmatrix} \delta_2 & e^{i\alpha_2} - 1 \\ \delta_3 & e^{i\alpha_3} - 1 \end{vmatrix}}{\begin{vmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 \end{vmatrix}} = \frac{-2.91987 - 0.35121i}{-0.5822 - 0.91168i} = 1.7264 - 2.10032i$$

$$R_3 = \frac{\begin{vmatrix} e^{i\beta_2} - 1 & \delta_2 \\ e^{i\beta_3} - 1 & \delta_3 \end{vmatrix}}{\begin{vmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 \end{vmatrix}} = \frac{0.18976 + 1.4195i}{-0.5822 - 0.91168i} = -0.2050 + 0.7722i$$

Figure 12: Analytical Synthesis of Four Bar Foot Mechanism - Standard Dyad Method

6.0 Automation Design

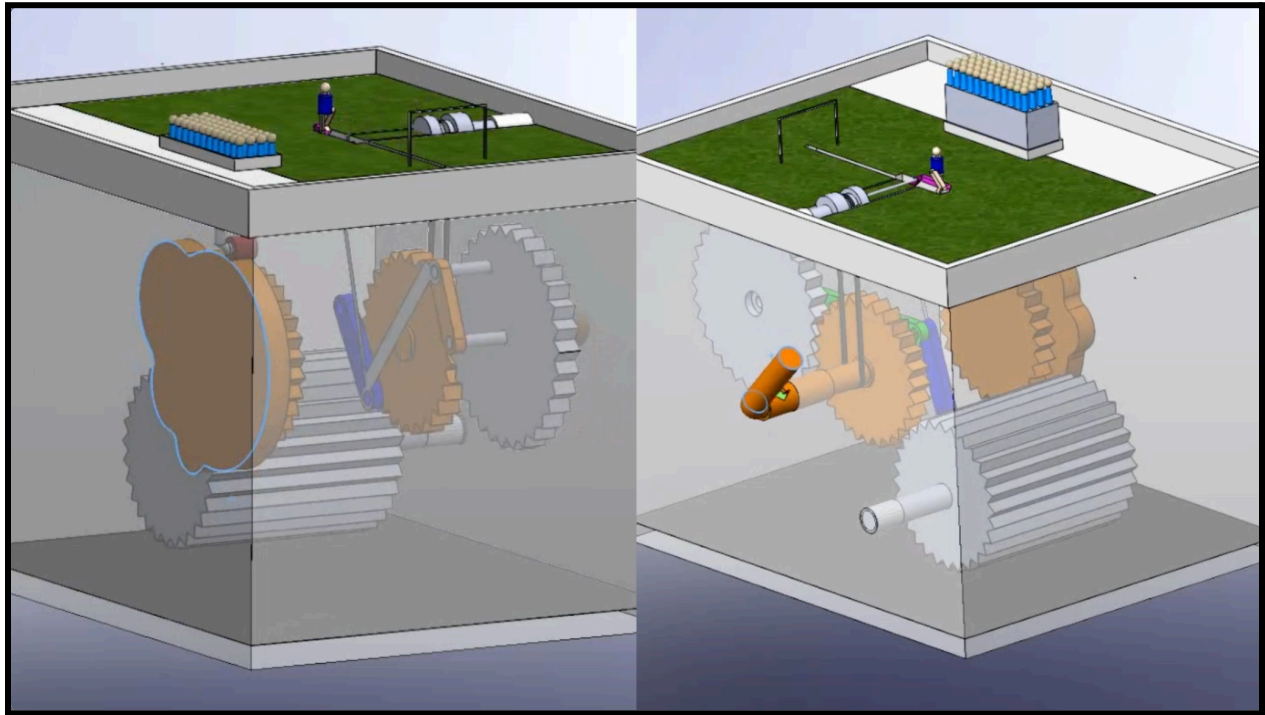


Figure 13: SolidWorks Automation Design

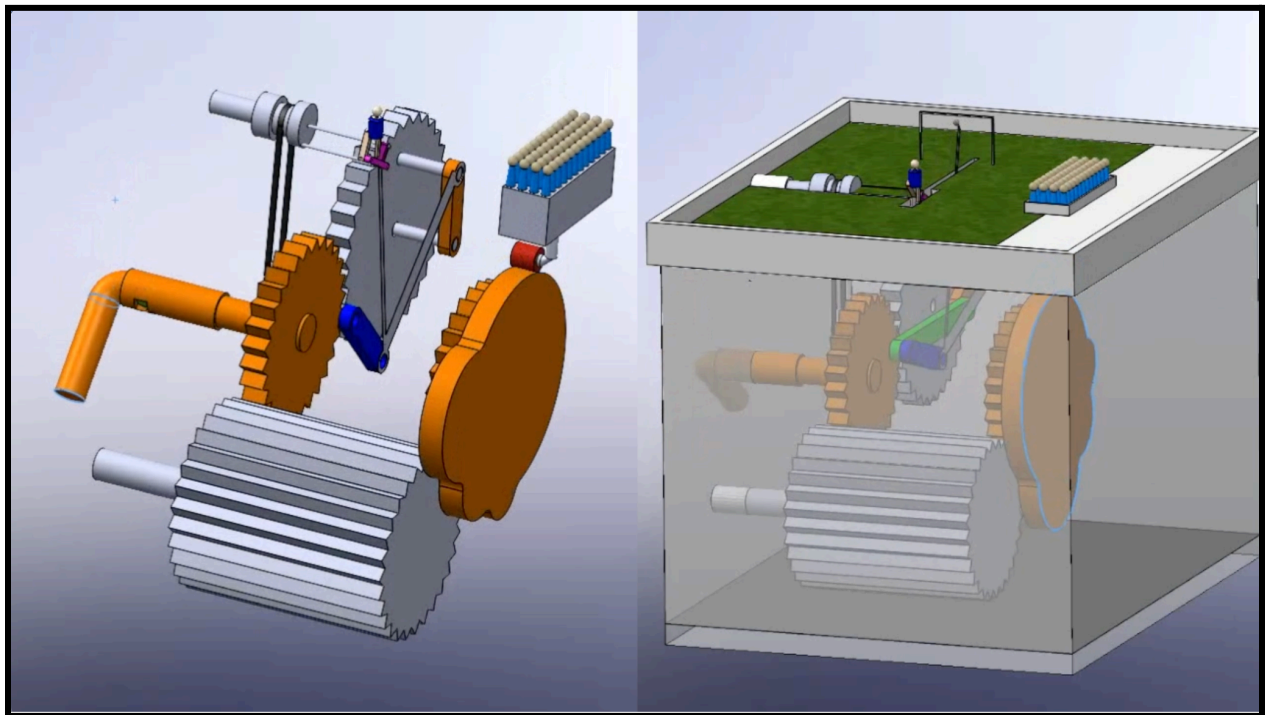


Figure 14: SolidWorks Automation Design

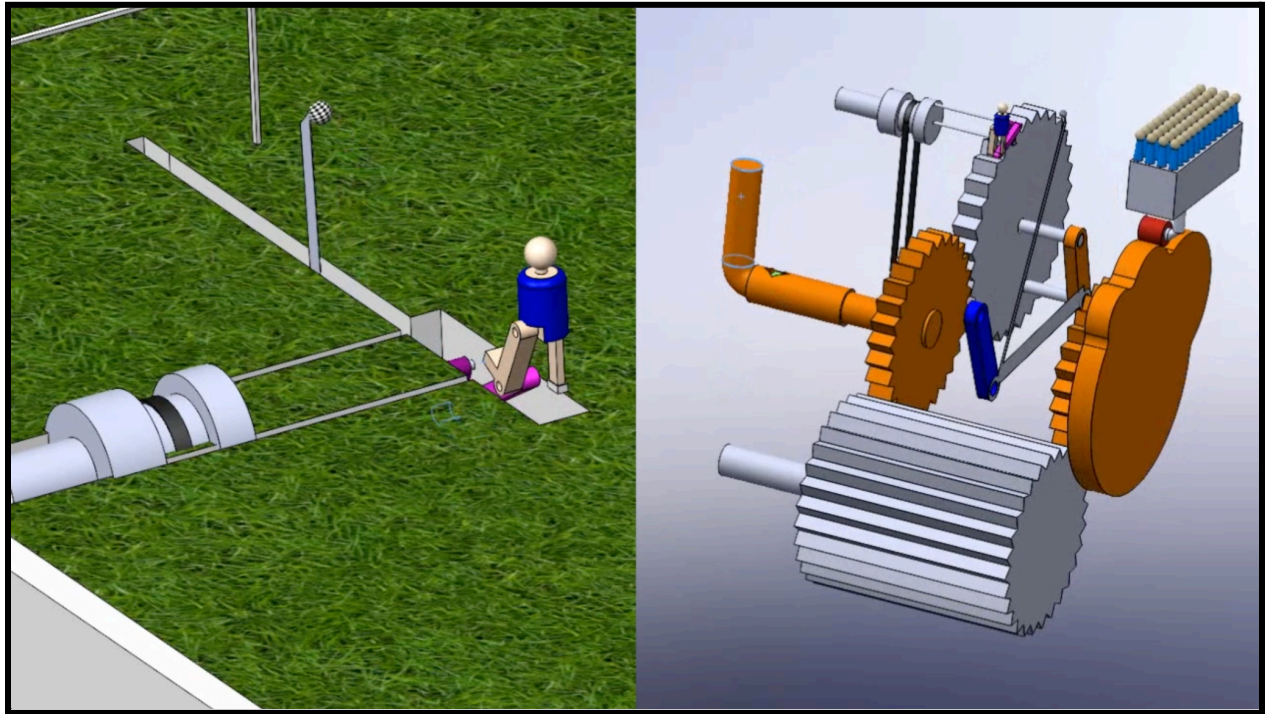


Figure 15: SolidWorks Automation Design

7.0 Working Model

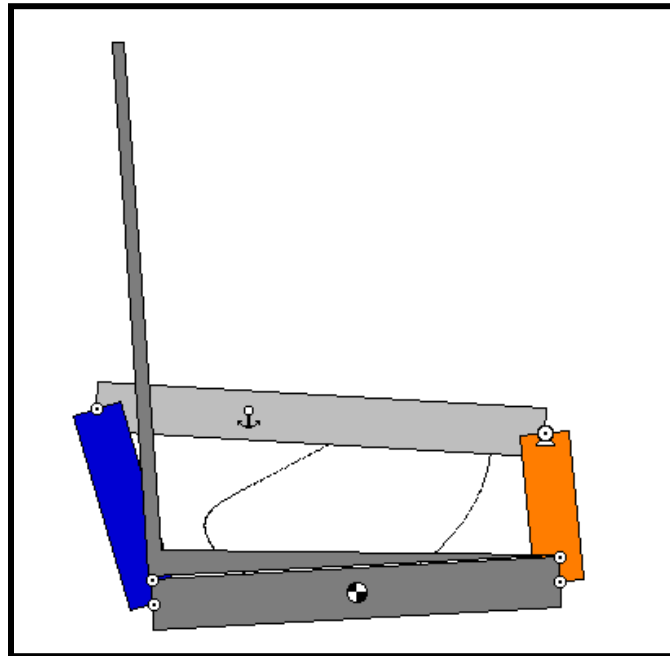


Figure 16: Working Model of Ball Mechanism and Coupler Curve

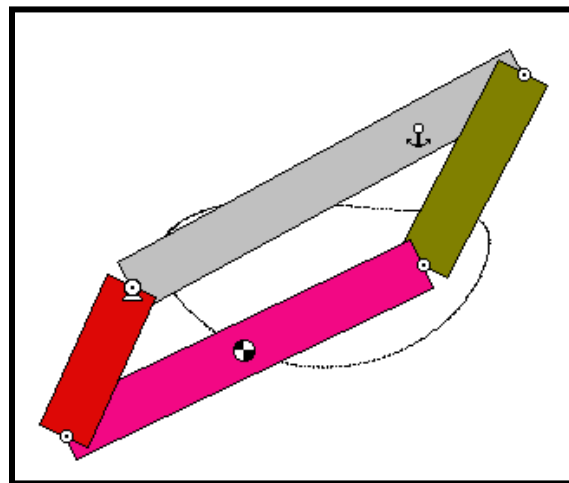


Figure 17: Working Model of Foot Mechanism and Coupler Curve

8.0 Design Reflection

The 'World Cup Goal' automaton, while successful in its depiction of a key sporting moment, presents opportunities for design enhancement. Reflecting on our design, we consider the following improvements:

1. **Optimization of Movement Trajectories:** Fine-tuning the linkage dimensions could lead to a smoother and more accurate representation of the soccer player's kicking motion and the ball's flight path. By refining the four-bar linkage geometries and cam profiles through iterative testing and simulations, we can achieve more fluid and lifelike movements.
2. **Enhancement of Mechanism Synchronization:** While gears improved the timing and coordination of the automaton's components, further refinement in gear ratios could be explored to optimize synchronization. We can experiment with different gear configurations to fine-tune the simultaneous movements of the ball and the crowd's reaction, ensuring that the celebration aligns precisely with the ball hitting the net.
3. **Improved Mechanism Integration:** In the current design, each mechanism functions independently within the system. We can improve the overall functionality by integrating the mechanisms more closely, allowing for a more compact design and reducing the complexity of the assembly. This integration can be achieved by designing multi-functional components that serve as both structural supports and active elements of the motion transfer process.

During the design process, we faced significant challenges with our initial choice of mechanisms. Our original design incorporated a spatial cylinder cam mechanism to drive the entire mechanism. However, this proved to be inefficient due to the complex profile required for the desired motion, and synchronizing the timing was particularly challenging. Similarly, the Scotch Yoke mechanism initially intended to animate the audience's movement encountered issues. The reciprocating motion of the yoke made it difficult to achieve the desired level of control and synchronization with the other elements of the automaton.

To resolve these issues, we transitioned to a gear-driven system. Gears provided us with a more reliable and straightforward method for transferring motion, allowing for easier adjustments and more precise timing during the construction phase in CAD. This modification simplified the design and assembly process while ensuring that the movements of the soccer player, ball, and audience were seamlessly coordinated.

9.0 Conclusion

In completing the automaton project, we successfully merged the principles of mechanical design with the creativity required to simulate a memorable sports event. This project pushed our engineering design abilities and collaborative skills, resulting in a functional mechanical design that brings the excitement of a soccer goal to life.

The project's core objectives—to design an automaton with at least three animated components, to synthesize two distinct types of 4-bar linkages, and to employ additional families of mechanisms—were met with a rigorous design that demonstrates both the complexity and the synthesis of mechanical mechanisms. Through the use of gears, linkages, and a cam mechanism, we achieved a harmony of movements that together tell the story of a memorable goal.

Throughout the project, we faced and overcame significant design challenges. Transitioning from a spatial cylinder cam to a gear-driven system improved our automaton's efficiency and synchronization, revealing the importance of adaptability and problem-solving in mechanical design. Our experiences have provided valuable lessons in the intricacies of mechanism design and the critical nature of component selection and timing